



Gigapixel camera by Duke University

A resolution that would highlight the smallest detail in the picture like a sky shot that captures a man walking way beneath, or a tear drop waiting to hit the ground!

Researchers at Duke University are developing wide-field, video-rate, gigapixel cameras in small, low-cost form factors. They are using multiscale designs that combine a monocentric objective lens with arrays of secondary microcameras. Following the multiscale lens design methodology, the field-of-view (FOV) is increased by arraying microcameras along the focal surface of the objective. Each micro-camera operates independently, offering much more flexibility in image capture, exposure and focus parameters.

Currently, a two-gigapixel prototype camera has been built. This system is capable of a 120-degree circular FOV with 226 microcameras, 38 microradian

FOV for a single pixel, and an effective f-number of 2.17. Each micro-camera operates at 10 fps at full resolution.

Displays made from air and water

An international team of researchers led by Aalto University has come up with an entirely new concept of writing and displaying information on surfaces using simply water. They have exploited the unique way a trapped layer of air behaves on a lotus-inspired, dual-structured, water-repelling surface immersed under water.

To achieve the extreme water-repellence of the lotus leaf, a surface needs to be superhydrophobic: It must have microscopic surface structures that prevent water from wetting the surface completely, leaving a thin layer of air between water and the surface. When such a surface is immersed in water, a trapped

air layer covers the entire surface.

The researchers led by Dr Robin Ras fabricated a surface with structures in two size scales: microposts that have a size of ten micrometres and tiny nanofilaments that are grown on the posts. On such a two-level surface the air layer can exist in two different shapes (wetting states) that correspond to the two size scales. The researchers found that one can easily switch between the two states locally using a nozzle to create over- or under-pressure in the water, in order to change the air layer to either state.

"The minimal energy needed to switch between the states means the system is bistable, which is the essential property of memory devices, for example," Dr Ras pointed out. There is a striking optical contrast between the states due to a change in the roughness of the water-air interface. Combined with the optical effect, the surface acts as a bistable reflective display.

Device makes electric currents visible

Engineers will now be able to see where electric currents are flowing in a circuit. For instance, when they charge a battery, using a magnetic imaging device they can see where the electricity accumulates inside.

The device has been developed by researchers from Kobe University's Graduate School of Science, led by Associate Professor Kenji Kimura. It gives a picture showing where electric currents are flowing in a circuit. In some cases, the researchers claim to have actually converted the images into measurements.

When a battery has been used for a long time, it ages, and it can't be charged anymore. This device shows where inside the battery is damaged

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